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Assignment - I

1)

Department	Employee	Salary
IT	A	50000
IT	B	60000
HR	C	45000
HR	D	55000
IT	E	70000
HR	F	65000

To display data frame:

```
import Pandas as Pd
data = {
        "Department" : ["IT", "IT", "HR", "HR", "IT", "HR"],
        "employee"   : ["A", "B", "C", "D", "E", "F"],
        "salary"     : [50000, 60000, 45000, 55000, 70000, 65000]
       }
df = Pd.dataFrame(data)
Print ("Employee data")
Print (df)
```

2) Pivot table to calculate average salary

```
import Pandas as Pd
data = { "Department", "employee", "Salary" }
  df => Pd.dataframe(data)
Pivot_avg => Pd.pivot_table( df, values = "salary",
                             index = "Department",
                             aggfunc = "mean"
                           )
Print ("Average Salary per department")
```

Print(Pivot_avg)

output:

Department
HR 55000
IT 60000

Pivot table to calculate total Salary for eac department

```
import Pandas as Pd
df = Pd.dataframe(data)
Pivot_Sum = Pd.Pivot_table(df, values = "Salary",
                           Index = "department",
                           aggfunc = "Sum"
                           )
Print ("total salary by department")
Print (df)
```

output:

department
HR 165000
IT 180000

④ Count Number of employees in each department

```
Pivot_Count = Pd.pivot_table( df, Values = "employee",
                              index = "Department",
                              aggfunc = "Count"
                              )
Print ("employee Count")
Print (df)
```

output:
department
HR 3
IT 3

5. max, min salary for each department

```
Pivot_max_min = pd.pivot_table(df, values="salary",
                               index="department",
                               aggfunc=["max","min"]
                               )

Print(Pivot_max_min
```

output:

department	max	min
HR	65000	45000
IT	70000	50000

6. multiple aggregation functions:

```
Pivot_all = pd.pivot_table(df, values="salary",
                           index="department",
                           aggfunc=["mean", "sum",
                                    "count", "max", "min"]
                           )

Print(pivot_all)
```

Scenario Question-1

1. customer age VS Purchase amount

```
import Pandas as pd
data = {
```

```
"customer": ["c1", "c2", "c3", "c4", "c5"],
"Age" : [20, 20, 25, 20, 30]
"Purchase Amount": [1500, 2000, 3000, 2500, 3500]
}
df = Pd.dataframe(data)
Pivot = Pd. Pivot_table(df, values = "Purchase Amount",
                        Index = "Age",
                        aggfunc = "Mean"
                        )
Print (Pivot)
```

Scenario question-2

① Sample dataframe

```
import Pandas as Pd
data = {
    "Department": ["IT", "IT", "HR", "HR", "Sales", "IT", "IT",
                   "HR", "Sales", "Finance"],
    "Gender" : ["M", "F", "M", "M", "M", "F", "M", "F", "F",
                "F"],
    "Salary" : [45000, 50000, 55000, 45000, 57000, 60000,
                65000, 53000, 90000, 98000]
}
df = Pd.dataframe(data)
Print (df)
```

ii) Average Salary

Avg_salary = Pd.Pivot_table(df, values = "salary",
index = "department",
aggfunc = "Mean"
)

Print (avg_salary)

iii) Total Salary by gender

gender_salary = Pd.Pivot_table (df, values = "Salary",
index = "gender",
aggfunc = "sum"
)

Print (gender_salary)

iv) Max, Min Salary

max_min = Pd. Pivot_table (df, Value = "Salary",
index = "department",
aggfunc = ["max", "min"]
)

Print (max_min)

Employee Count by department

count = Pd. Pivot_table (df, Values = "salary",
index = "department",
aggfunc = "count"
)

Print (count)